

A Dual-Expert Routing Framework for Structured Data Extraction from Scientific Tables

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ABSTRACT

Despite extensive research, extracting structured data from scientific tables remains a challenging problem due to merged cells, hierarchical headers, mathematical notation, and domain-specific symbols. Recently, multimodal language models have significantly improved layout analysis via the alignment between visual and textual content. However, experiments on complex tables show that vision-language models falter on dense symbolic content, while text-based language models preserve fine-grained notation but mishandle complex structure. Prior work treats these two paradigms as competing alternatives; we use this failure as an informative signal. We present a dual-expert framework that runs a vision expert and a text expert over the table image, aligns their cell predictions into "decision units" via Hungarian matching on grid-*IoU*, and trains a lightweight per-cell router to select the more reliable expert's output for each cell. Across the two datasets, A25 and SciTSR, and four backbone models (GPT-4o, Gemini 3.1, Gemma 4, Qwen 3.6), the router improves per-cell extraction accuracy over the best single expert in all eight configurations, with an average gain of 2.48%. These results show that a simple learned arbitrator can recover part of the complementary signal between experts, pointing toward a design principle for multimodal table extraction systems. Our code and data are available at: <https://anonymous.4open.science/r/dual-expert-table-extraction-9B60>

CCS CONCEPTS

• **Computing methodologies** → **Ensemble methods**; **Natural language processing**; • **Information systems** → **Information extraction**.

KEYWORDS

Scientific Table Data Extraction; Large Language Models; Vision-Language Models

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1 INTRODUCTION

In scientific literature, tables serve as a primary medium for presenting structured data, including experimental results, comparative statistics, and observations [9, 19]. Moreover, extracting these tables into machine-readable formats supports applications such as scientific search and retrieval [9], knowledge base construction [7, 10], and automated evidence synthesis [19, 25]. However, tables embedded in PDF documents remain among the most challenging document elements to extract, due to complex layouts and dense semantic content [13, 20], as well as spanning rows and columns, hierarchical headers, mathematical expressions, and domain-specific notation [1].

Two paradigms have emerged to improve table data extraction: vision-based approaches that handle layout and structure well but struggle with dense symbolic content, and text/LaTeX-based approaches that capture notation faithfully but often fail on structural complexity such as merged and spanning cells [25?]. Recent work has begun to combine these paradigms, for example by routing table elements to specialized experts within a unified model [29] or selecting between visual and textual inputs at the document or table level, but routing decisions at the level of individual cells, between two independent experts, remain underexplored.

Ground Truth	Vision Expert	Text Expert
<i>ta</i> 1.0	<i>ta</i> 1.0	<i>ta</i> 1.0
<i>tios</i> [†]	<i>tios</i> +	<i>tios</i> [†]

Table 1: Examples demonstrating the complementary behavior of the two experts. The vision expert reads the rendered table image, while the text expert parses Nougat-transcribed LaTeX markup and renders each cell's content into plain text. Errors are highlighted in red. These comparisons were taken from Gemma 4 as the LLM backbone.

A key observation motivates our work: the two experts fail on different subsets of cells, so an oracle that selects the correct expert for each cell has the potential to outperform either single modality. The gap between oracle routing and either single-expert baseline is therefore an underutilized source of information, showing that jointly leveraging both experts' capabilities could yield gains.

To close this gap, we present a dual-expert framework that pairs a vision expert with a text expert for structured, cell-level data extraction from scientific tables. We establish an oracle routing ceiling that quantifies the headroom available to any cell-level router, train a router on cell-level features derived purely from the two experts' predictions, and measure how much of the gap between the single-expert baselines and the oracle it closes. Finally, we test the robustness of this result across five machine learning models and two large language models as the router families and four backbone LLM models.

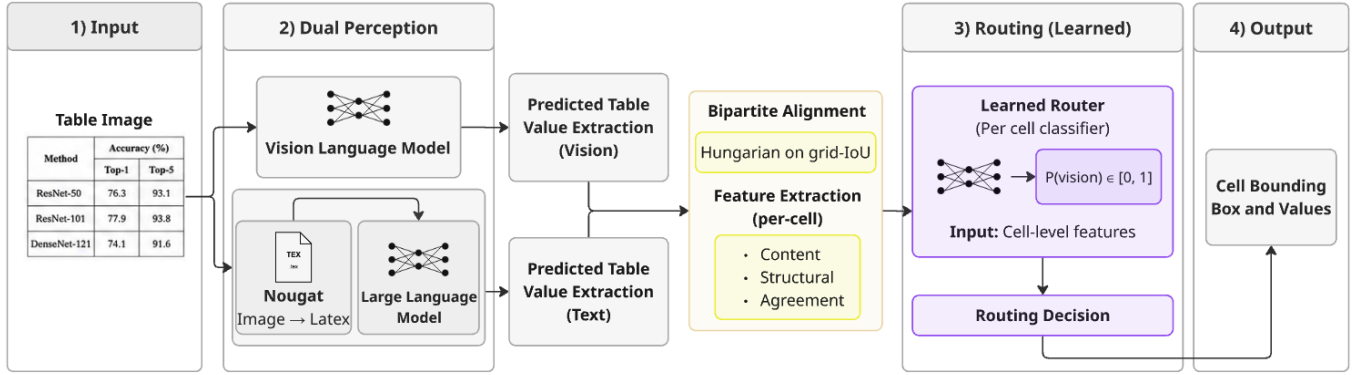


Figure 1: Overview of our dual-expert framework for structured data extraction of scientific tables

2 BACKGROUND AND RELATED WORK

Early table extraction methods relied on handcrafted rules and heuristics [20]. However, modern pipelines typically separate table structure recognition (TSR) from text extraction (OCR) [13]. Benchmarks such as SciTSR [6], PubTabNet [33], PubTables-1M [24], and TabLeX [13] have driven progress, with performance commonly evaluated using precision, recall, F1, and TEDS [3, 33]. Nevertheless, structured data extraction from complex scientific tables remains challenging due to merged cells, mathematical expressions, domain-specific symbols, and structure recognition errors that often propagate to later extraction stages [2].

More recently, end-to-end models such as Donut [15], Pix2Struct [17], and Nougat [4], as well as multimodal large language models, generate structured representations such as Markdown, LaTeX, or JSON directly from images without a separate TSR-OCR pipeline [18, 32]. These vision-based models are often effective on complex layouts, including merged cells and hierarchical headers [14], but can struggle with dense numerical data, equations, and specialized scientific notation [27]. Text-based approaches, in contrast, preserve fine-grained symbol characters more accurately but recover table structure less reliably [30].

One way to leverage both modalities is to fuse them within a single model. Systems such as LayoutLMv3 [12] and UDOP [26] jointly encode text, images, and document layout into a shared representation, and UniTabNeT [31] extends this idea to table structure recognition. However, since these models produce a single prediction from a single fused representation, they cannot tell when one modality is more reliable than the other, or reason about cases where the visual and textual evidence disagree.

An alternative direction keeps specialized models separate and learns when to trust each of them [5, 11, 28]. For example, Hybrid LLM [8] and RouteLLM [22] estimate query difficulty and route an entire input to the most appropriate model. Closest to our work, ComProScanner [23] and nanoMINER [21] coordinate vision-based and text-based agents to extract from scientific articles, but they operate at the document level and target scattered entities rather than complete table structures.

Our approach follows this expert paradigm but adapts it to structured data extraction from scientific tables, where different modalities exhibit different strengths between regions of the same table. This motivates routing at a finer, per-cell granularity, combining

the structural strengths of vision models with the textual fidelity of text-based models so that each is used where it is most reliable [1].

3 METHODOLOGY

Our framework, as shown in Figure 1, treats structured data extraction of scientific tables as a classification problem between two complementary extraction experts. Given a table image, a vision expert and a text expert independently generate structured table representations. A lightweight router then determines, at the cell level, which expert’s prediction should be trusted.

To enable direct comparison between the two experts, both produce outputs in a common cell-based schema. This schema is built on a grid system where rows and columns are indexed starting from zero. Each cell is represented as a tuple $(sr, er, sc, ec, text)$, where sr (start row) and er (end row) define the vertical span of the cell, sc (start column) and ec (end column) define the horizontal span of the cell, and $text$ contains the extracted cell content.

Vision and text expert. The vision expert directly extracts table cells from the table image using a vision-language model (VLM). The model is constrained to generate outputs that conform to the predefined cell schema, allowing both structural and textual information to be produced in a single step. The text expert follows a two-step process. First, the table image is converted into LaTeX markup using Nougat [4]. The resulting markup is then provided to a large language model (LLM), which converts the LaTeX representation into the same cell schema. To improve structural consistency, we adopted a few-shot in-context learning method by providing examples that cover common scientific-table challenges such as multirow and multicolumn spans, mathematical expressions, and Greek symbols. Figure 2 illustrates one of the few-shot examples used in our prompt. Both experts are decoded deterministically using a temperature of zero to ensure reproducible outputs.

Prediction alignment. Although both experts generate the same schema, they frequently disagree on table structure. For example, one agent may predict a spanning cell while the other splits the same region into multiple cells. As a result, cells cannot be compared solely by row and column indices. To address this issue, we align the predictions of the two experts and ensure that the alignment procedure remains identical during both training and inference. For each pair of cells (v, t) , where v is the vision cell and t is the

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### Example 3: Combined Multirow/Multicolumn + Math Symbol Errors
BAD LaTeX INPUT:
\begin{tabular}{|c|c|c|}
\hline
\multirow{2}{*}{\beta-phase} & \multicolumn{2}{c}{Properties} \\
& \sigma_y \text{ (MPa)} & E \text{ (GPa)} \\
\hline
Ti-6Al-4V & 924 \text{ \pm 15} & 114 \text{ \pm 2} \\
\end{tabular}

WRONG OUTPUT:
[{"text": "\beta-phase", "start_row": 0, "end_row": 0, "start_col": 0, "end_col": 0},
{"text": "Properties", "start_row": 0, "end_row": 0, "start_col": 1, "end_col": 1}]

WHAT'S WRONG: 1) \multirow{2} span ignored, 2) \multicolumn{2} span wrong,
3) Math symbols not converted

CORRECT OUTPUT:
[{"text": "\beta-phase", "start_row": 0, "end_row": 1, "start_col": 0, "end_col": 0},
{"text": "Properties", "start_row": 0, "end_row": 0, "start_col": 1, "end_col": 2},
{"text": "\sigma_y \text{ (MPa)}", "start_row": 1, "end_row": 1, "start_col": 1, "end_col": 1},
{"text": "E \text{ (GPa)}", "start_row": 1, "end_row": 1, "start_col": 2, "end_col": 2},
{"text": "Ti-6Al-4V", "start_row": 2, "end_row": 2, "start_col": 0, "end_col": 0},
{"text": "924 \pm 15", "start_row": 2, "end_row": 2, "start_col": 1, "end_col": 1},
{"text": "114 \pm 2", "start_row": 2, "end_row": 2, "start_col": 2, "end_col": 2}]

```

Figure 2: A few-shot example given to the text expert

text cell, we compute an alignment score based primarily on the overlap between their table footprints. Spatial overlap is measured using intersection-over-union (IoU), while normalized Levenshtein similarity between cell contents is used as a secondary signal for tie-breaking among overlapping candidates. We then solve a one-to-one assignment problem using the Hungarian algorithm [16]. Matched pairs whose IoU falls below a threshold are discarded and treated as unmatched cells. The resulting alignment produces a set of decision units of the form (v, t) , $(v, \emptyset)$, or $(\emptyset, t)$.

Per-cell routing. Each decision unit is evaluated by a routing model that predicts which expert’s extraction is more likely to be correct for that cell. The router is built using features derived solely from the experts’ predictions, including textual agreement, cell length, numerical and symbolic content indicators, span geometry, and the presence or absence of a corresponding prediction from either expert. The router is trained only on disagreement cases where the two experts produce different outcomes and exactly one prediction matches the ground truth. These examples provide explicit supervision regarding which expert should be trusted. The selected cells are then combined to produce the final table extraction.

4 EVALUATION

To evaluate our framework, we used two complementary datasets. The first is the A25 dataset from the paper on uncertainty-aware complex scientific table extraction [2], which contains tables from four domains: Biology, Computer Science, ICDAR, and Materials Science. The second is a representative subset of the SciTSR corpus [6]. Since running both experts across four backbone models is computationally expensive, evaluating the full SciTSR corpus of 15,000 tables was infeasible. Due to this computational cost, we limited our SciTSR evaluation to a subset of only 157 tables in total.

Table 2: Statistical features of the two datasets.

Metric	SciTSR	A25
#Tables	157	156
#Cells	9,287	9,620
Rows (mean)	9.7	11.2
Rows (max)	30	43
Cols (mean)	6.7	5.6
Cols (max)	17	14
Tables w/ multirow cell	37.6%	18.6%
Tables w/ multicolumn cell	86.0%	30.8%

Rather than sampling this subset at random, we started from the 716 tables that SciTSR designates as “complicated” in its provided annotations. Of these 716 labeled complicated tables, 666 produced usable text-expert output, and from this pool we randomly selected our final 157 tables. Table 2 reports the resulting statistics. We count multi-row and multi-column cells as separate per-table indicators, because a single table may contain both kinds of spanning cells.

To ensure a rigorous evaluation and eliminate data leakage, we implemented strict table-level partitioning for both datasets. Since the A25 dataset does not have a predefined split and the SciTSR data was sampled, we partitioned the data into train, validation, and test splits. We use a 60/15/25 train/validation/test split, and all reported results are computed on the held-out test tables only.

To establish robustness, we instantiate the vision and text experts using four distinct backbone models: two proprietary models (GPT-4o and Gemini 3.1) and two open-source models (Gemma 4 and Qwen 3.6). For each backbone, both experts are executed, followed by alignment via bipartite matching.

A predicted cell is deemed correct if and only if it satisfies a strict dual-criterion: its span (start and end rows/columns) must exactly match a ground-truth cell, and its normalized textual content must agree. Textual agreement is quantified using the normalized Levenshtein similarity [2]:

$$\text{Levenshtein_Accuracy}(gt, pred) = 1 - \frac{\text{Levenshtein}(gt, pred)}{\max(\text{len}(gt), \text{len}(pred))}$$

where $\text{Levenshtein}(gt, pred)$ is the character-level edit distance between the two strings. We enforce an exact match threshold of $\tau = 1.0$, requiring perfect textual agreement after standard normalization (e.g., whitespace stripping and Unicode math symbol alignment). We report the following metrics to evaluate the performance for each configuration:

- **Single Expert:** Standalone accuracy of an individual expert.
- **Router:** The accuracy of our dual-modality system
- **Oracle:** The ceiling of an optimal selector that picks the correct expert whenever either is right
- **Gain:** The absolute percentage point (pp) improvement of the router over the stronger of the two single modalities.

To select the optimal router, we evaluated five distinct machine learning models: a Random Forest (RF), Gradient Boosting (GB), a Multi-Layer Perceptron (MLP), Logistic Regression (LR), and a Support Vector Machine (SVM) with a radial basis function (RBF) kernel. We then ranked the models based on the gain they produced

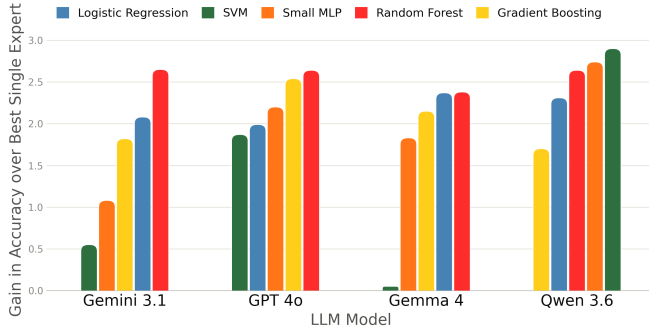


Figure 3: Performance comparison of different routing models on the A25 Dataset

Table 3: Average rank of five router models across the eight dataset and backbone configurations (lower is better).

Routing Model	RF	Small MLP	GB	LR	SVM
Average Ranking	1.62	2.75	3.00	3.38	4.25

over the best single expert across all eight configurations (assigning a rank of 1 to the highest gain).

As illustrated in Figures 3 and 4, all router families yield positive gains in nearly all configurations, confirming that the performance gain is intrinsic to the dual-expert framework rather than a fortuitous choice of classifier. However, the tree-based ensemble method, Random Forest, consistently achieved the top ranks, with the Random Forest attaining the best average rank of 1.62. Consequently, we adopt the Random Forest as the standard ML router to test the robustness of our results across backbone models.

In addition to the five feature-based ML routers, we tested whether a large language model can perform the routing itself. In this variant, the cell predictions of both experts were provided as JSON in the prompt of a separate arbiter large language model, which is instructed to decide between the two readings cell by cell and emit the final table directly in the same cell schema, with no explicit alignment step. We evaluated two arbiter models: GLM 5.2, selected as a representative open-weight model, and GPT 5.6, selected as a leading proprietary model, allowing us to compare routing performance across both model classes. Despite their far greater capacity, both LLM routers trail the Random Forest in most configurations, suggesting a lightweight ML router attends to the routing signal more reliably than a general-purpose LLM arbiter.

The consistent gain achieved by most of the ML routers in all eight configurations, as shown in Table 4, supports our central claim: a dual-expert system with a lightweight per-cell router improves per-cell extraction accuracy over the best standalone expert. Specifically, the Random Forest (RF) router exceeds the best single expert in all eight configurations, yielding absolute gains ranging from 2.05% to 3.04%.

Moreover, the four backbones differ substantially in capability and modality bias; nevertheless, as shown in Figure 3 and 4, routing yields a positive gain for each configuration. This indicates that the benefit derives substantially from the complementarity of visual and textual readings rather than from any single model. Moreover,

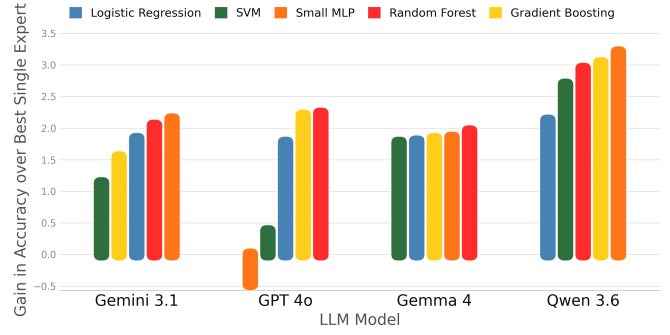


Figure 4: Performance comparison of different routing models on the SciTSR dataset

Table 4: Robustness of the accuracy of dual-expert framework for data extraction. The Gain column reports the absolute percentage point improvement of the Random Forest (RF) router over the best of the two single expert baselines.

Dataset	Model	Single		Router			Oracle	Gain
		Vision	Text	RF	GLM 5.2	GPT 5.6	(ceiling)	
SciTSR	GPT-4o	69.98	81.94	84.28	80.48	83.04	86.17	2.33
	Gemini 3.1	77.55	82.24	84.38	83.21	83.09	85.89	2.14
	Gemma 4	84.91	86.38	88.43	87.54	87.02	89.80	2.05
	Qwen 3.6	67.99	81.06	84.10	87.54	87.30	88.40	3.04
A25	GPT-4o	65.36	80.44	83.08	75.26	78.85	85.65	2.64
	Gemini 3.1	73.56	80.34	82.99	81.53	79.06	85.17	2.65
	Gemma 4	78.22	83.14	85.52	78.93	84.53	91.23	2.38
	Qwen 3.6	78.96	81.44	84.08	85.17	82.29	87.45	2.64

the absolute gain remains stable at approximately two percentage points even as the oracle gap varies widely, suggesting the router captures many of the complementary signal from local, cell-level features.

5 CONCLUSIONS AND FUTURE WORK

We presented a dual-expert framework for structured data extraction from scientific tables that treats the disagreement between a vision expert and text expert as signal to be exploited. Rather than fusing the two readings into a single prediction or committing to one model for an entire table, the framework aligns the extractors' cells into decision units and routes between the two experts to extract the valid spans and values for the cells. This per-cell router consistently improves extraction accuracy over the best single expert across all eight configurations, with Random Forest proving the most reliable router. The main cost is running both experts per table; the router itself adds negligible overhead. Moreover, it closes only part of the headroom the oracle reveals, pointing to two avenues for future work: letting the router attend to a cell's row, column, and local neighborhood through graph-based context, since table errors are rarely independent; and extending the framework beyond two experts and beyond the backbones studied here to widen the space of complementary signal available. We leave a fuller exploration of these directions, and of the remaining oracle gap, to future work.

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